Sloan School of Management
Massachusetts Institute of Technology

Andrew W. Lo
Paul F. Mende

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Modeling with Machine Learning: Financial Technology

Project #2

Choose one of the topics below and develop an appropriate ML solution. Explain your choice of model, training data, and testing procedures.

Projects will be evaluated according to these criteria:

- Completeness of the financial analysis (as compared to similar reports published by major investment banks) (25 points)
- Novelty of the analysis (25 points)
- Readability and attractiveness of the report (25 points)
- Completeness of source attribution, including all LLMs used and the specific prompts involved (25 points)

Project Topics

- **Quantamental Analysis:**
  - Using the last 10 years of public filings of Nvidia, perform a detailed fundamental analysis of the stock and provide a recommendation for investors regarding an investment in the company, i.e., is it a "buy", "hold", or "sell" and why. Include a detailed analysis of how scalable your approach is, i.e., how easy would it be to automate the generation of your report?
- **Credit Analysis:**
  - Using credit approval data from the UCI Machine Learning Repository, develop and backtest an ML model for credit scoring that yields some form of explainability for individual credit decisions.
- **Technical Analysis:**
  - Use an ML image classification pipeline to evaluate images of asset price patterns and predict future price movements. Evaluate its effectiveness based on statistical criteria and investment criteria.
- **Build a Better Beta:**
  - The notion of beta – the sensitivity of a security to market moves – is widely used but poorly controlled. Devise an ML model to predict future realized beta without any of the parametric assumptions of the CAPM or its extensions. Analyze its performance in portfolio risk management and corporate finance.
- **Build a Better Stress Test:**

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- Build a scenario engine that can generate extreme but realistic scenarios using ML models (diffusion model, GAN, VAE, ...) trained on market data and conditioned on triggering events or regimes. Compare its outputs to out-of-sample historical events. Is it realistic? Does it hallucinate? Is it suitable for risk management or trading?

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